

exif-turbo User Manual

exif-turbo lets you scan image folders, build a searchable index of all EXIF metadata, and instantly find any photo by camera model, lens, date, location, or any other tag — across thousands of images.

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1. Requirements

ExifTool

exif-turbo requires **ExifTool** on your `PATH` to extract metadata from images.

Platform	Install
Windows	Download the standalone <code>.exe</code> from exiftool.org , rename to <code>exiftool.exe</code> , place in a folder on your PATH
macOS	<code>brew install exiftool</code>

Linux	<code>sudo apt install libimage-exiftool-perl</code>
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2. Installation

Windows installer (recommended)

Download `exif-turbo-<version>-windows.msi` from the [Releases page](#). The installer adds an entry to **Start Menu** and puts `exif-turbo` on your `PATH`.

From source

```
pip install -e .
```

Command-line options

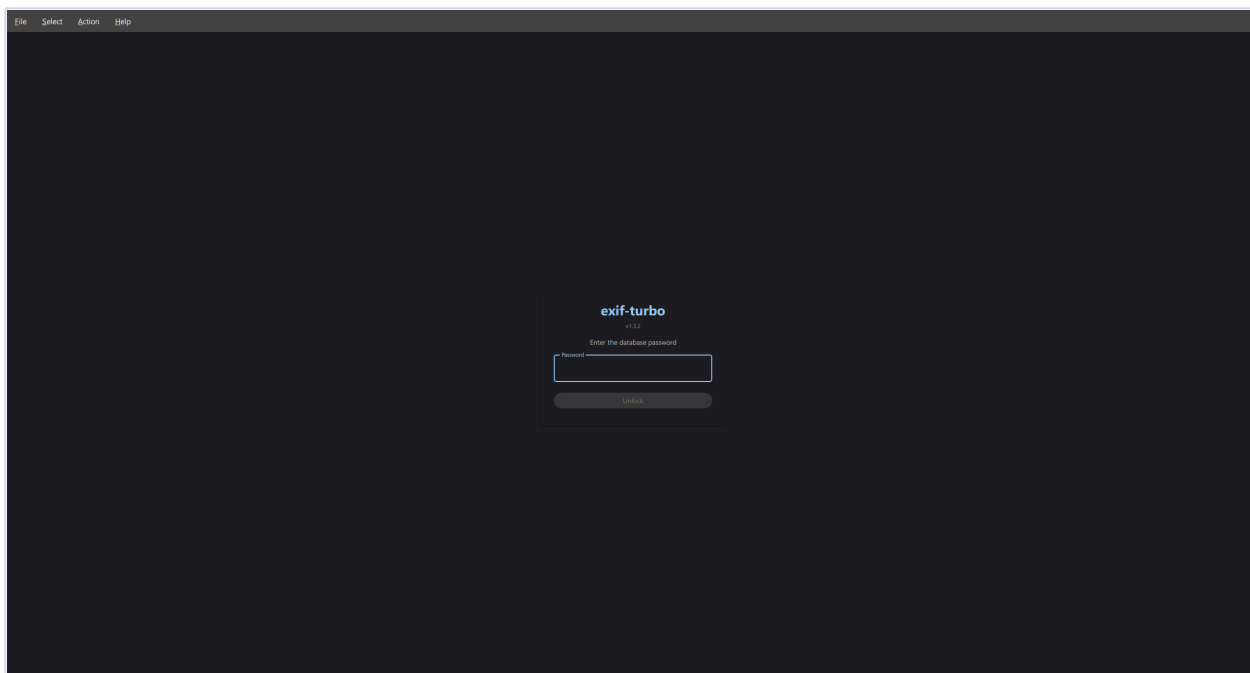
```
exif-turbo [--db NAME]
```

Option	Description
<code>--db NAME</code>	Open (or create) a named database instead of the default one. The database is always stored in <code>~/.exif-turbo/data/<NAME>/<NAME>.db</code> . Useful for keeping separate libraries — e.g. <code>exif-turbo --db work</code> and <code>exif-turbo --db holidays</code> .

If `--db` is omitted, the default database `~/.exif-turbo/data/index/index.db` is used.

3. First Launch — Unlocking the Database

When you start `exif-turbo` you are greeted by the **lock screen**:



First-time setup — creating a new database

If this is the first time you have launched exif-turbo (or you are using a new named database), the lock screen shows a **New passphrase** field, a **Confirm passphrase** field, and a **Create Database** button. Choose a strong passphrase of at least 12 characters that mixes letters, numbers, and symbols — it encrypts your entire image index and **cannot be recovered if lost**.

Opening an existing database

Enter your password in the **Password** field and click **Unlock** (or press **Enter**). The same password is required every time you open the app.

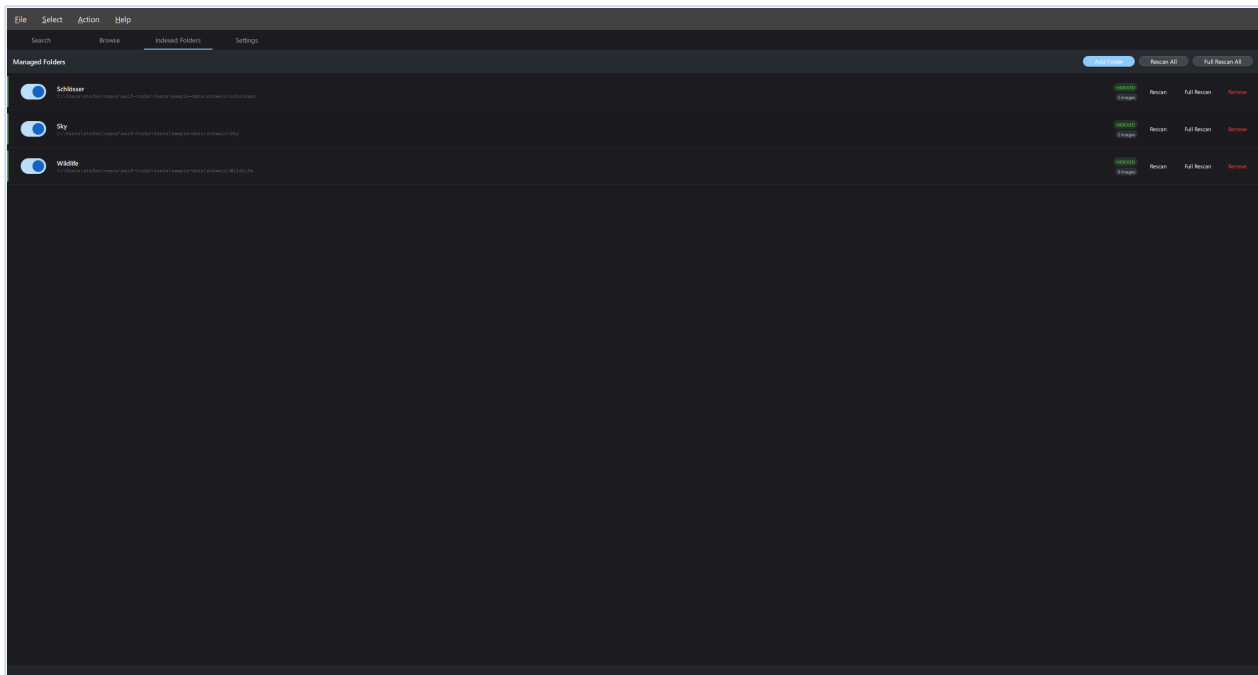
Once unlocked, the **Search** tab opens and any previously indexed images are immediately available.

Help menu

The **Help** menu in the menu bar provides access to this user manual, the third-party open-source licence list, and the **About** dialog, which shows the application version, a brief description of the application, and the licence (MIT).

4. Indexed Folders — Managing Your Library

Click the **Indexed Folders** tab to manage which directories are scanned.



Adding a folder

1. Click **Add Folder** (top-right of the Indexed Folders tab).
2. Pick the directory in the file browser dialog.
3. The folder is immediately queued for scanning — its status changes to **QUEUED** then **SCANNING** once the worker starts.

Starting an index scan

Click **Rescan** next to a folder (or **Rescan All** to queue all enabled folders). The folder status changes to **SCANNING** and the progress panel appears in the bottom-right corner of the tab.

Use **Full Rescan** (or **Full Rescan All**) to force every file to be re-processed even if its modification time has not changed. This is useful after updating ExifTool or if you suspect the index is out of date.

When rescanning a single folder, only the records belonging to that folder are updated or removed. Images indexed from other folders are not affected.

Folder statuses

Each folder row shows a coloured status badge and an image count badge (e.g. "42 images") once it has been indexed.

Status	Meaning
NEW	Added but never scanned

QUEUED	Waiting in the scan queue
SCANNING	Currently being indexed
INDEXED	Last scan completed successfully
MISSING	Folder path no longer exists on disk
ERROR	Last scan ended with an error
DISABLED	Excluded from search results

Disabling / enabling a folder

Toggle the **Enabled** switch to exclude a folder from search results without deleting it or its index data.

Removing a folder

Click **Remove**. A confirmation dialog asks you to confirm before the folder and all its indexed images are deleted from the database. The original files on disk are not touched.

5. Indexing Progress

While scanning, a non-blocking progress panel appears in the bottom-right corner of the **Indexed Folders** tab:

- **Queue indicator** — shows which folder is currently being scanned. When multiple folders are queued it reads "**Indexing folder 2 of 5**"; when only one folder is being processed it reads "**Indexing**".
- **Progress bar** — shows an indeterminate animation while exif-turbo is scanning the folder for image files, then switches to a percentage once the total file count is known.
- **File counter** — shows "**Scanning for images...**" during the initial discovery phase, then **n / total** files once counting is complete.
- **Current file** — name of the file being processed.
- **Cancel Indexing** button — stops indexing immediately. While the worker is stopping the label changes to "**Canceling...**" and the button is disabled until the worker has stopped. During the thumbnail-build phase the same button shows **Cancel Thumbnails**.

The same progress panel is reused after indexing finishes to show **thumbnail building** progress ("Building Thumbnails"). Thumbnails are generated in a background pass and cached to disk so

subsequent launches are fast.

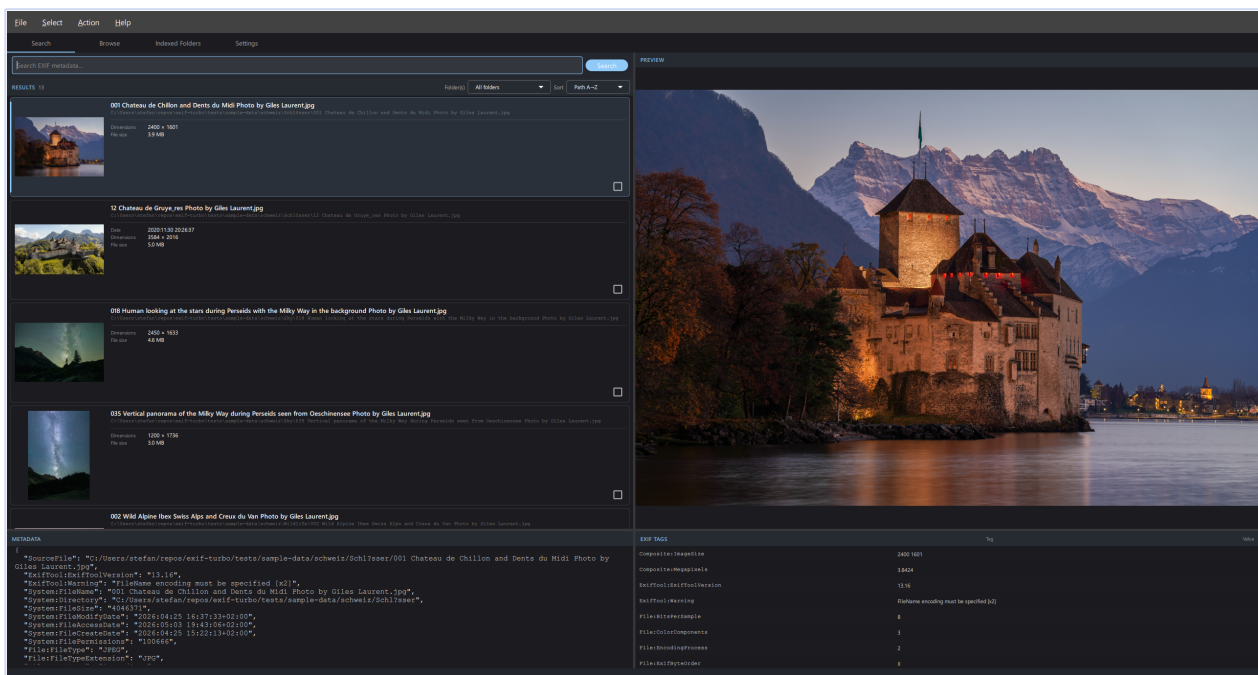
Across **all tabs** the **status bar** at the very bottom of the window shows a pulsing blue dot and the text **Indexing...** during the file-indexing phase, so you always know the indexer is running even when you are working in Search or Browse. The dot is not shown during the separate thumbnail-building phase. The status bar also shows brief event messages to its right (such as "Indexed 42 images" after a scan completes).

Pause and resume

If you close the application while indexing is in progress, the current folder is saved as **QUEUED**. The next time you open exif-turbo and unlock the database the scan queue is automatically restored and indexing resumes where it left off.

6. Searching

The **Search** tab is the main way to find photos:



Photos: © [Giles Laurent](#), CC BY-SA 4.0

Running a search

Type any word or phrase into the **Search EXIF metadata...** bar and press **Enter** (or click **Search**). exif-turbo performs a full-text search across every metadata field — camera make and model, lens name, date, GPS coordinates, keywords, copyright, and more.

Press **Enter** with an empty search bar to show all indexed images.

Filtering by format

When results contain more than one file format, a row of format chips appears below the search bar. Click a chip to show only that format:

All	CR2 · 1 459	JPG · 563	TIF · 113	PNG · 2
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Click **All** to return to unfiltered results.

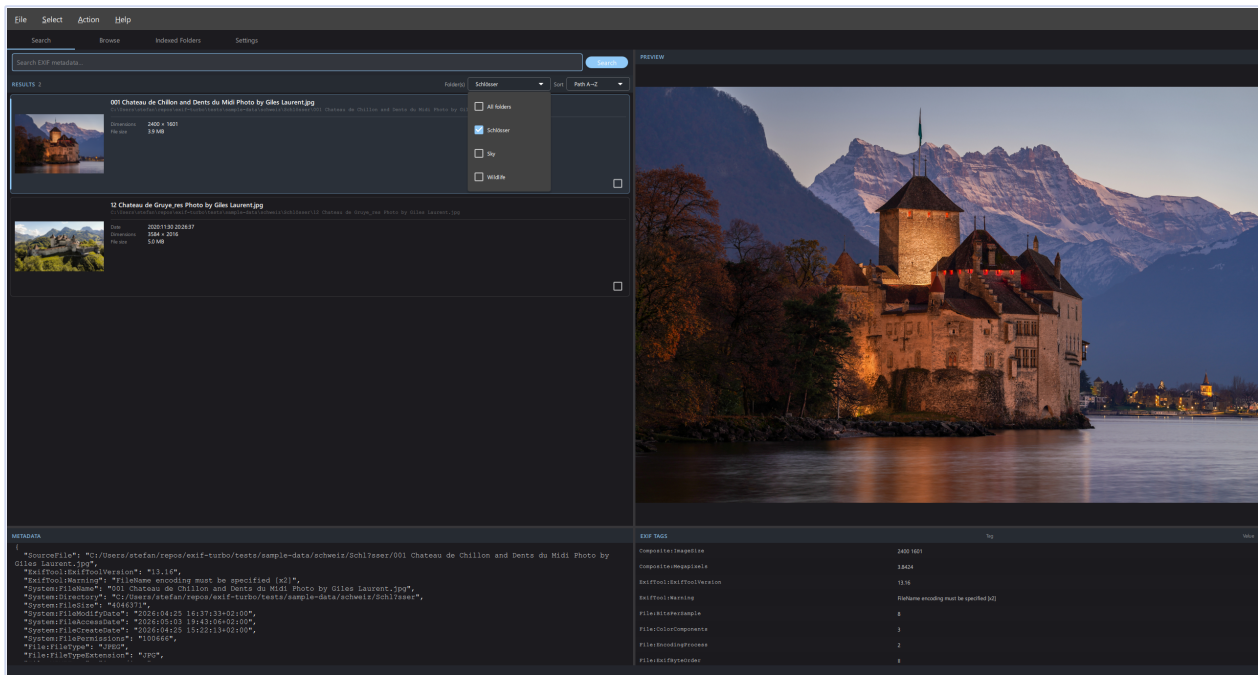
Filtering by folder

When you have two or more indexed folders enabled, a **Folder(s)** dropdown appears in the RESULTS header bar, to the left of the **Sort** control. It is hidden when only one folder is indexed.

The button label reflects the current filter state:

Label	Meaning
All folders	No folder filter — results span all enabled folders
Folder name	Exactly one folder is selected
N folders	N individual folders are selected

Click **Folder(s)** to open the selection popup. Tick **All folders** to clear any active filter; tick one or more folder names to restrict results to those folders only. Multiple folders can be selected simultaneously. Hovering over a folder name shows its full path as a tooltip.



Photos: © [Giles Laurent](#), CC BY-SA 4.0

Sorting results

Use the **Sort** dropdown at the top-right of the results panel:

Option	Description
Name A→Z	Filename ascending
Name Z→A	Filename descending
Path A→Z	Full path ascending (default)
Path Z→A	Full path descending
Newest first	Date taken, most recent first
Oldest first	Date taken, oldest first
Largest	File size, largest first

Loading more results

Results are loaded in batches as you scroll. When you reach the bottom of the result list the next batch is fetched automatically. The total match count is shown next to the **RESULTS** badge in the panel header.

Opening images from results

Single-click a result card to select it and load the preview and metadata panels.

Double-click a result card to open the file or folder in your system's default application: -
Double-clicking the **thumbnail** (left side of the card) opens the image in your default image viewer. -
Double-clicking the **info area** (right side of the card) opens the parent folder in the system file manager. On Windows, Explorer opens with the file highlighted; on macOS and Linux the parent folder is opened.

Search examples

Query	What it finds
Canon EOS R5	All images shot with a Canon EOS R5
f/1.4	All images taken at f/1.4 aperture
Switzerland 2024	Images with Switzerland and 2024 in any metadata field
eagle	Images whose filename, title, or keywords mention eagle
Nikon Z 9 ISO 6400	Nikon Z 9 shots at ISO 6400

Advanced query syntax

The search box accepts the full **SQLite FTS5** query language:

Syntax	Example	What it does
term	Canon	Keyword anywhere in metadata
"exact phrase"	"red deer"	Terms must appear adjacent and in order
term1 AND term2	Canon AND 50mm	Both terms must be present
term1 OR term2	Canon OR Nikon	Either term must be present
term1 NOT term2	50mm NOT Nikon	First term present, second term absent

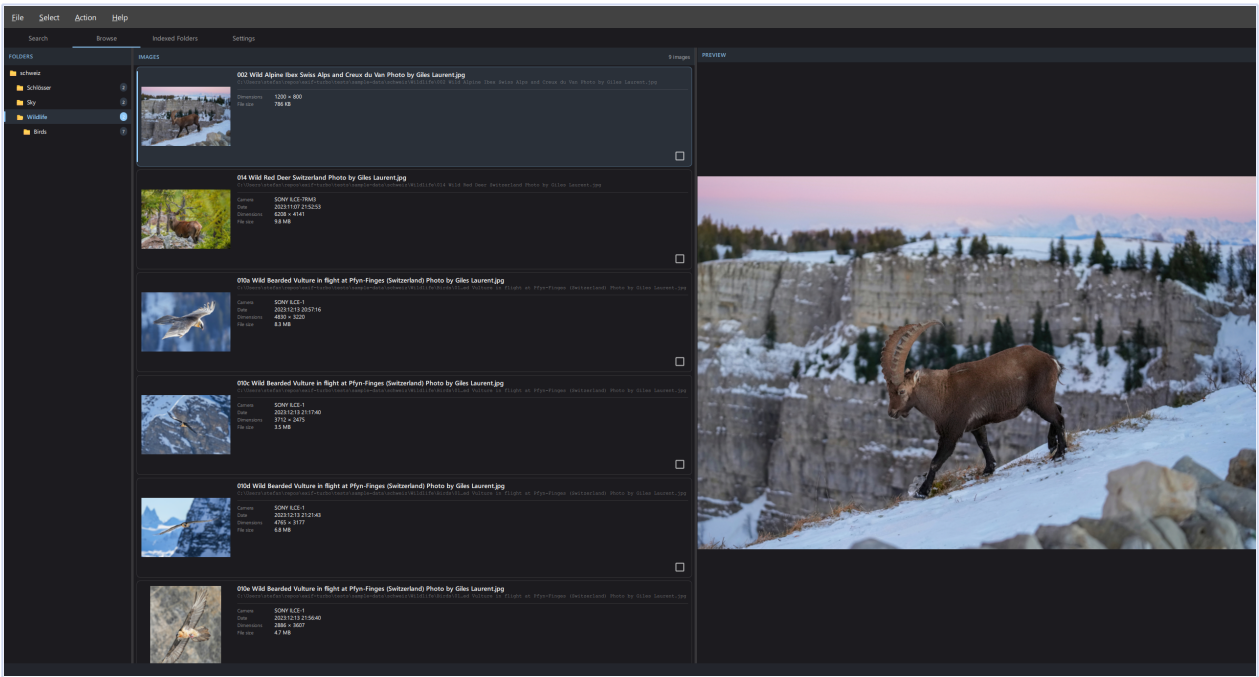
<code>column:term</code>	<code>filename:IMG_1234</code>	Search within a specific field only
<code>prefix*</code>	<code>Fuji*</code>	Matches any token starting with the prefix

Available column names for scoped queries: `path`, `filename`, `metadata_text`.

Multiple terms without an operator (`Canon 50mm`) are treated as an implicit AND.

7. Browsing by Folder

The **Browse** tab lets you navigate your library by folder hierarchy:



Photos: © [Giles Laurent](#), [CC BY-SA 4.0](#)

The left panel shows all indexed folders as an indented list — sub-folders are indented under their parent. Each entry shows the folder name and a count of images inside it. Click any folder to show only its images in the centre panel. Click the highlighted folder again to deselect it and show all images.

The same thumbnail list and preview pane appear as in Search. **Single-click** an image to select it and load the preview. The preview supports the same zoom and pan gestures as in Search (scroll wheel, touchpad scroll, touchpad pinch-to-zoom, drag-to-pan, double-tap to reset — see [Zooming and panning](#)). **Double-click** an image to open it in your system's default viewer. (Unlike the Search tab, double-clicking in Browse always opens the image — there is no folder-open shortcut.) The

Metadata and EXIF Tags panels are not shown in the Browse tab; use the **Search** tab for the full metadata view of a selected image.

Switching to the **Search** tab clears the folder filter and re-runs the current search query. Any image previously selected while browsing is not automatically highlighted in Search.

8. Viewing Metadata and EXIF Tags

Selecting any image in the result list populates three panels:

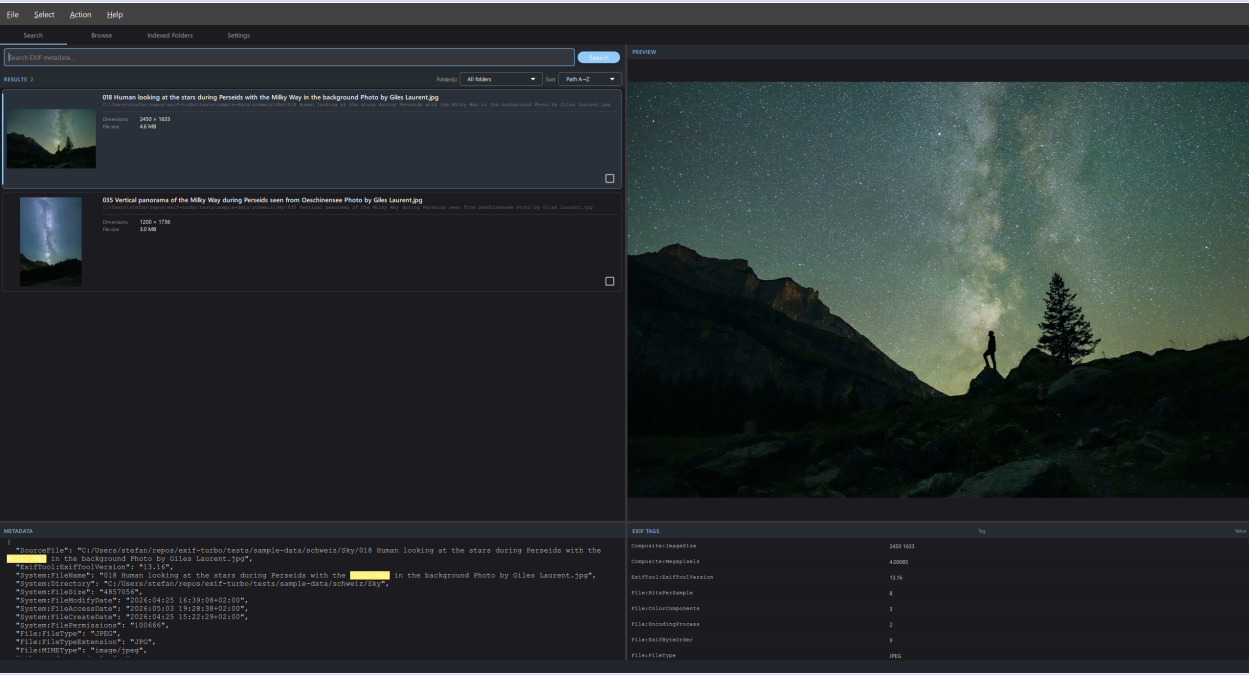


Photo: © [Giles Laurent](#), CC BY-SA 4.0

Preview

The right pane shows a full-resolution preview of the selected image scaled to fit the available space. For RAW files (CR2, CR3, NEF, ARW, DNG, etc.) the embedded preview JPEG is used. A cached thumbnail is shown immediately as a low-resolution placeholder while the full image is loading; the full image fades in once it has been decoded.

Zooming and panning

The preview supports cursor-anchored zoom and drag-to-pan:

Input	Action
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Scroll wheel — up / down	Zoom in / out, anchored to the cursor position
Mouse — click and drag	Pan the zoomed image in any direction
Touchpad — two-finger swipe up / down	Smooth zoom in / out, anchored to the cursor position
Touchpad — two-finger swipe left / right	Pan left / right
Touchpad — two-finger drag	Pan in any direction
Touchpad — pinch	Zoom in / out, anchored to the pinch centroid
Double-click / double-tap	Reset zoom to fit

Zoom is capped at **8×**. A badge in the bottom-right corner of the preview shows the current zoom level when it is above 1×. Selecting a new image always resets the zoom to fit.

Metadata panel (bottom-left)

Displays the metadata for the selected image as formatted, indented JSON. Click the **Find** button in the panel header (or press **Ctrl+F**) to open an inline search bar and find any tag value. Press **F3** (or click the ▼ / ▲ arrows) to jump through all matches. Search terms from the main search bar are highlighted automatically.

EXIF Tags panel (bottom-right)

Displays the same metadata as a clean two-column table — **Tag** and **Value** — sorted alphabetically. Hover over a truncated tag or value to see the full text in a tooltip.

You can drag the divider between the two bottom panels to adjust the split. By default they start at **50 % / 50 %**.

9. Settings

Click the **Settings** tab to configure application behaviour.

Worker Threads

Controls the number of parallel threads used for indexing and thumbnail generation. Higher values speed up processing on multi-core machines but use more CPU and memory. The default is half the

number of detected CPU threads.

Indexing Blacklist

A list of file and folder name patterns to skip during indexing. Supports wildcards: `*` matches any sequence of characters, `?` matches a single character. Patterns are matched against individual file or folder names, not full paths.

Examples: `@eaDir`, `*.tmp`, `Thumbs.db`

Changes to the blacklist take effect on the next rescan.

Theme

Choose between **system** (follows OS dark/light mode), **light**, or **dark**. The theme changes immediately.

Language

Select the display language from the dropdown. A restart is required for the language change to take full effect.

Change Password

Click **Change Password...** to re-encrypt the SQLCipher database under a new password. The dialog asks for:

- **Current password** — must match the password the database is currently encrypted with.
- **New password** — your new passphrase. Cannot be empty and must differ from the current one.
- **Confirm new password** — must match the new password exactly.

Click **Change Password** to apply. A busy indicator and the message “**Changing password... This may take a moment.**” are shown while exif-turbo re-encrypts every page of the database. Existing **thumbnails are preserved** — exif-turbo uses a wrapped-key model, so only the wrapping key (which protects the thumbnail key) is re-encrypted under the new password; the thumbnail cache on disk does not have to be rebuilt.

Once the operation finishes successfully the dialog closes and a confirmation appears. The new password is required the next time you unlock the database.

If the current password is wrong, an inline red error message (“Current password is incorrect.”) is shown and the dialog stays open. The **Change Password...** button is disabled while indexing or thumbnail building is in progress, and while the database is locked.

There is no recovery if you forget the new password. Make sure you remember it (or store it in a password manager) before clicking **Change Password**.

Reset Database

At the bottom of the Settings tab, a divider separates the standard settings from a destructive-action zone.

Click the **Reset Database...** button (red) to open a confirmation dialog. Click **OK** to confirm. This permanently:

- Deletes all indexed images from the database
- Removes all indexed folder records
- Wipes the thumbnail cache on disk

The database is vacuumed and checkpointed immediately, so the database file shrinks to near-zero on disk straight away.

This action cannot be undone. After a reset you will need to re-add your folders and run a full rescan to rebuild the index.

The **Reset Database...** button is disabled while indexing is in progress.

10. Keyboard Shortcuts

Shortcut	Action
Enter	Run search
Ctrl+F	Open / close find-in-metadata bar
Escape	Close the find-in-metadata bar
F3	Jump to next match in metadata
Shift+F3	Jump to previous match in metadata
↓	Select the next result (Search tab)
↑	Select the previous result (Search tab)

Page Down	Jump one page forward in results (Search tab)
Page Up	Jump one page backward in results (Search tab)
Ctrl+Q	Quit

11. FAQ

Q: Why does the status bar say "Indexing..." even after I switch tabs?

A: The indexer runs in the background across all tabs. The pulsing blue dot in the status bar lets you know it is still working. You can continue searching and browsing while indexing proceeds.

Q: I closed the app mid-scan. Will I lose my progress?

A: No. exif-turbo saves the queue state when it closes. Next time you unlock the database, any interrupted scans are automatically resumed.

Q: The search finds nothing even though I can see files in the folder.

A: The files must be indexed first. Go to the **Indexed Folders** tab, add the folder, and click **Rescan**.

Q: Does exif-turbo modify my image files?

A: Never. exif-turbo only *reads* metadata — it never writes to your images.

Q: What image formats are supported?

A: JPEG, PNG, TIFF, HEIC, BMP, GIF, and RAW formats: CR2, CR3, NEF, ARW, DNG, ORF, RW2, PEF, RAF, RWL, SRW (and any other format that ExifTool can read).

Q: Where is the database stored?

A: By default at `~/exif-turbo/data/index/index.db` on all platforms.

Q: How do I change the database password?

A: Open the **Settings** tab and click **Change Password....** Enter your current password and a new one — exif-turbo will re-encrypt the database in place. Thumbnails are preserved (no re-indexing required). See [Change Password](#) for details.

Q: Thumbnails are not showing / are slow to appear.

A: Thumbnails are generated in a background thread after indexing. Depending on the number of images it may take a few minutes for all thumbnails to be built and cached. They persist across sessions so subsequent launches are fast.

Image Credits

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